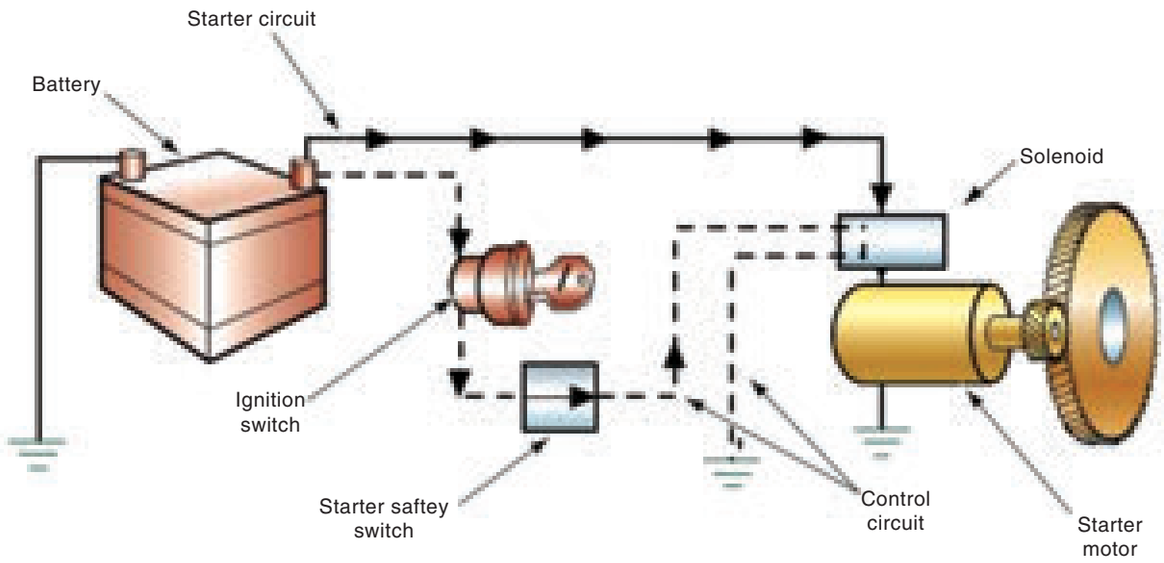


Figure 10.8 Engine starting system

10.8.1 Starter motor drive mechanism

The ratio between the starting motor pinion gear and the ring gear in engine flywheel is varied from 1:10 to 1:15. For example, if the ratio 1:15 means, the flywheel will rotate 1 revolution when the motor pinion rotates 15 revolutions. Once, the engine is cranked, then the speed of the engine will increase. If the starter motor is continuously engaged, then the engine will drive the motor and the speed of the motor will be 15 times higher than the engine speed. This will lead to failure of the starting motor. To prevent this, pinion gear should be disengaged from the flywheel ring gear once the engine is started and the mechanism used for disengagement is called as starter motor drive mechanism.

Types of starter motor drive mechanism

1. Bendix drive
2. Over running clutch type
3. Folo – thru system
4. Rubber compression system
5. Sliding armature system
6. Parallel type system
7. Friction clutch system

10.8.1.1 Bendix drive starting system

Construction

There is a threaded sleeve in an armature shaft. The sleeve will move on the shaft back and forth. Armature shaft is directly connected to the drivehead. Coil spring is mounted over the sleeve. Pinion will be over the sleeve. There is an unbalanced weight on the pinion.

Working

When the starter switch is "ON", armature and sleeve assembly rotate. While rotating, the pinion moves over the sleeve due to inertia as pinion is loosely mounted over the sleeve. The pinion is engaged with the flywheel ring gear and start the engine. As the engine speed increases, the pinion retracts and go away from the engine flywheel.

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